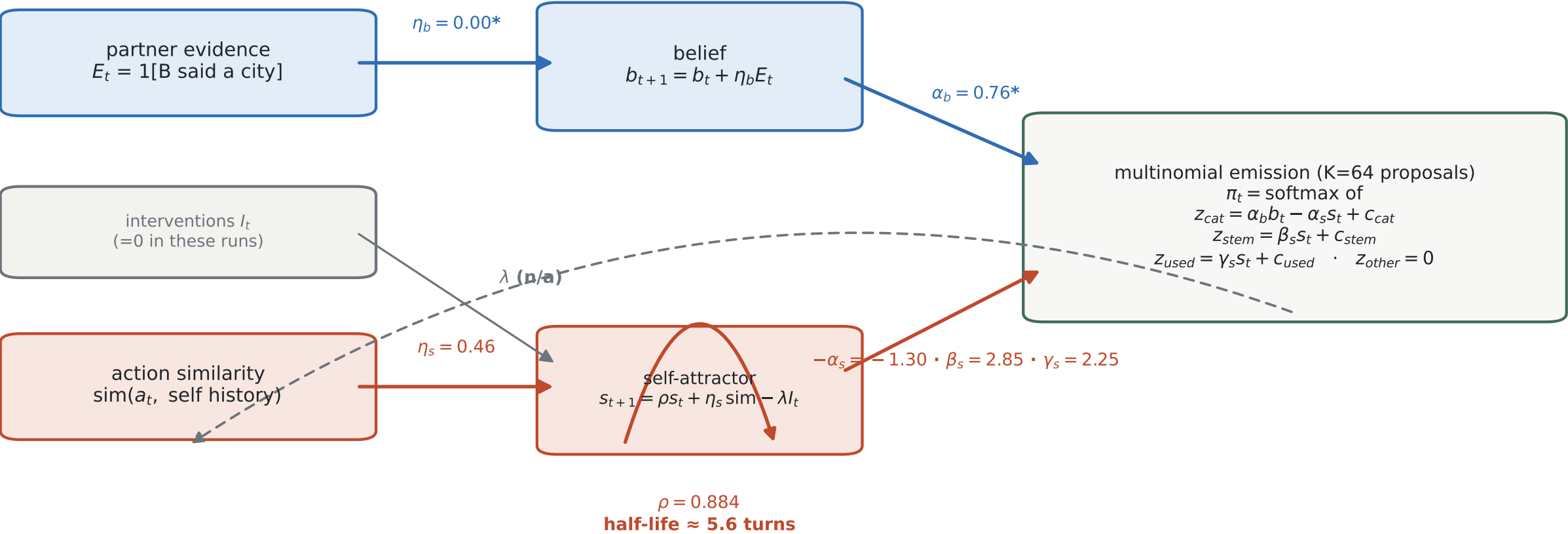


Two-process state-space model

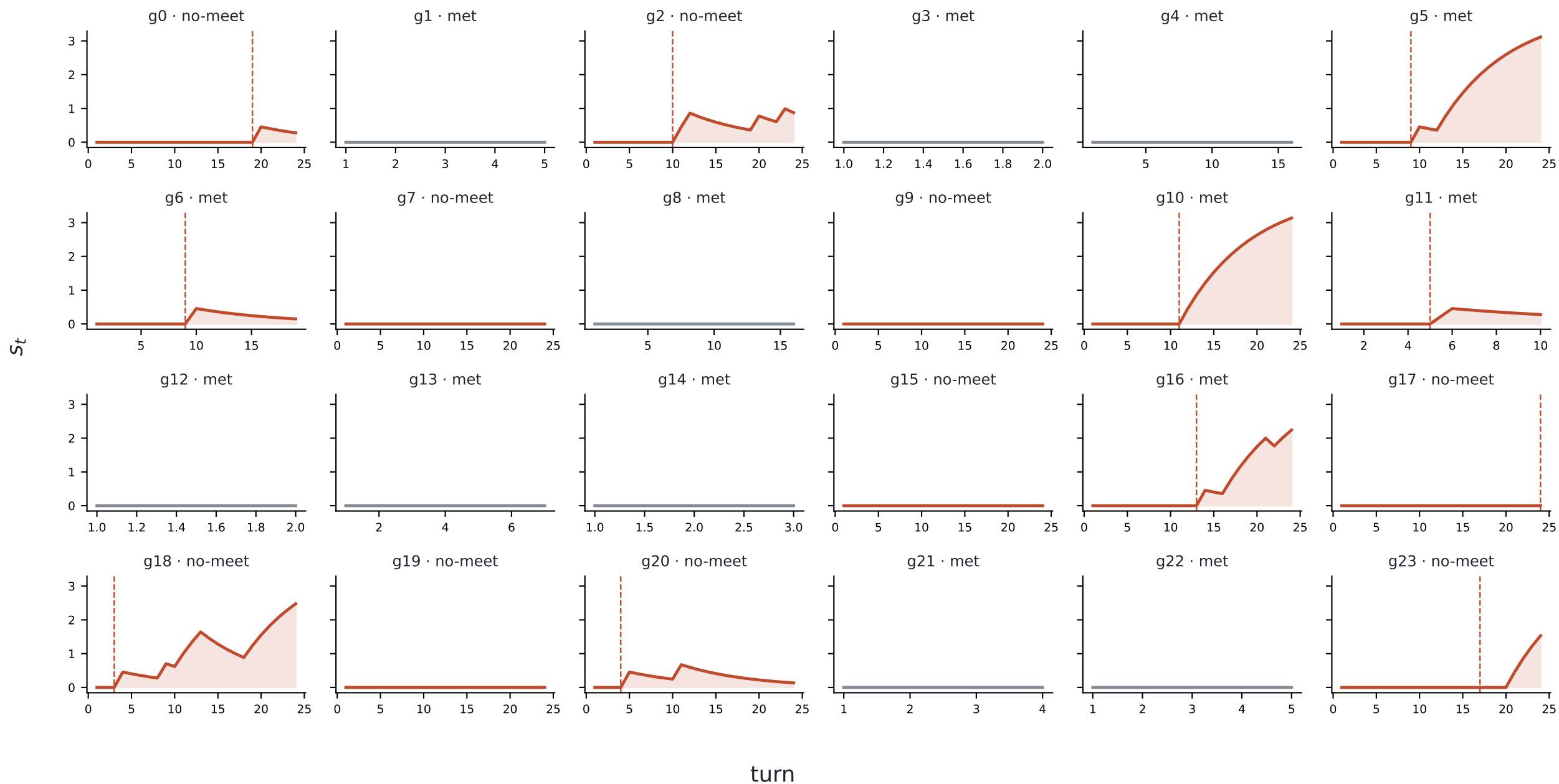
belief about the partner and a self-attractor compete for the proposal distribution



sampled action a_t feeds back into self-history $\rightarrow$ the loop that makes the attractor SELF-REINFORCING

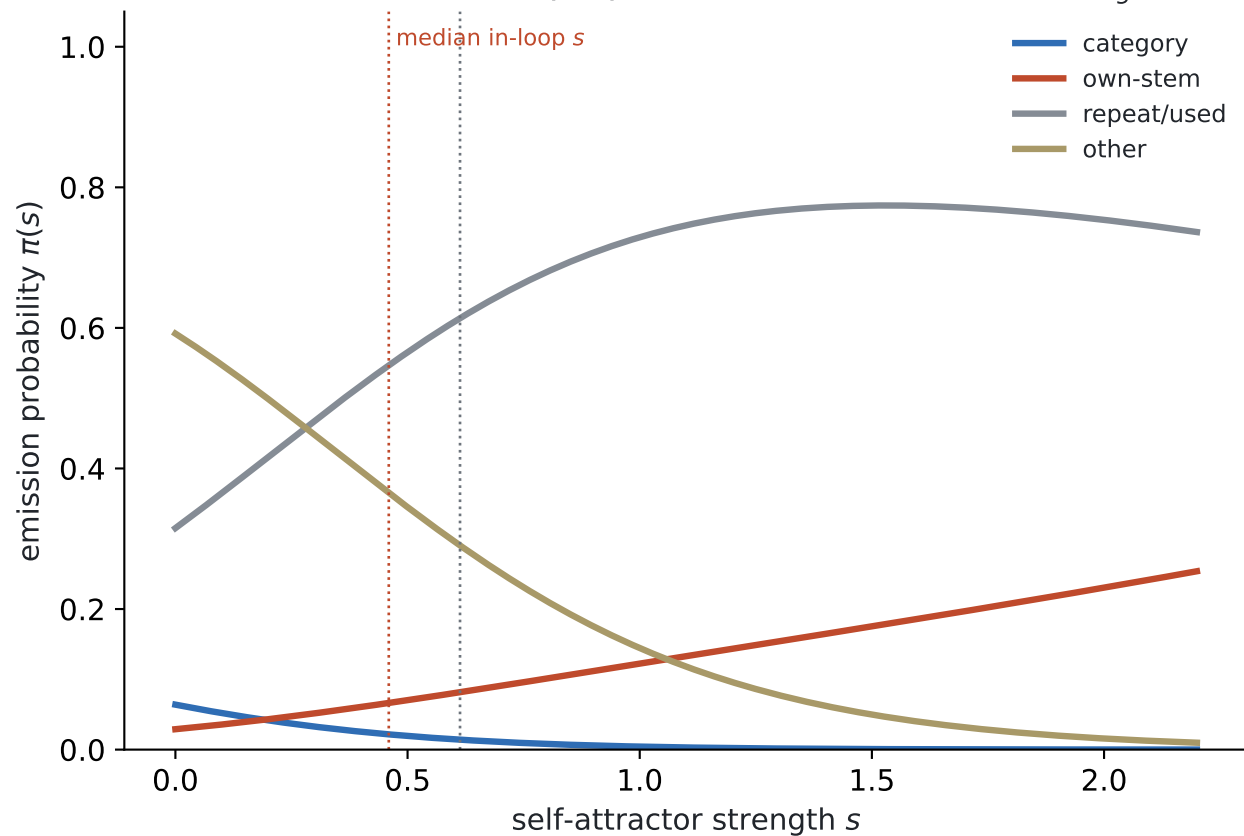
* $\eta_b \rightarrow 0$, α_b inert: belief is unidentified from action-channel emissions alone (category mass falls in onset games while forced-choice knowledge stays 0.98 — the dissociation). Constants c_{cat} , c_{stem} , c_{used} and c_{other} are set to 0.

Fitted self-attractor S_t per game (rust = onset games, grey = never-onset; dashed line = onset turn)

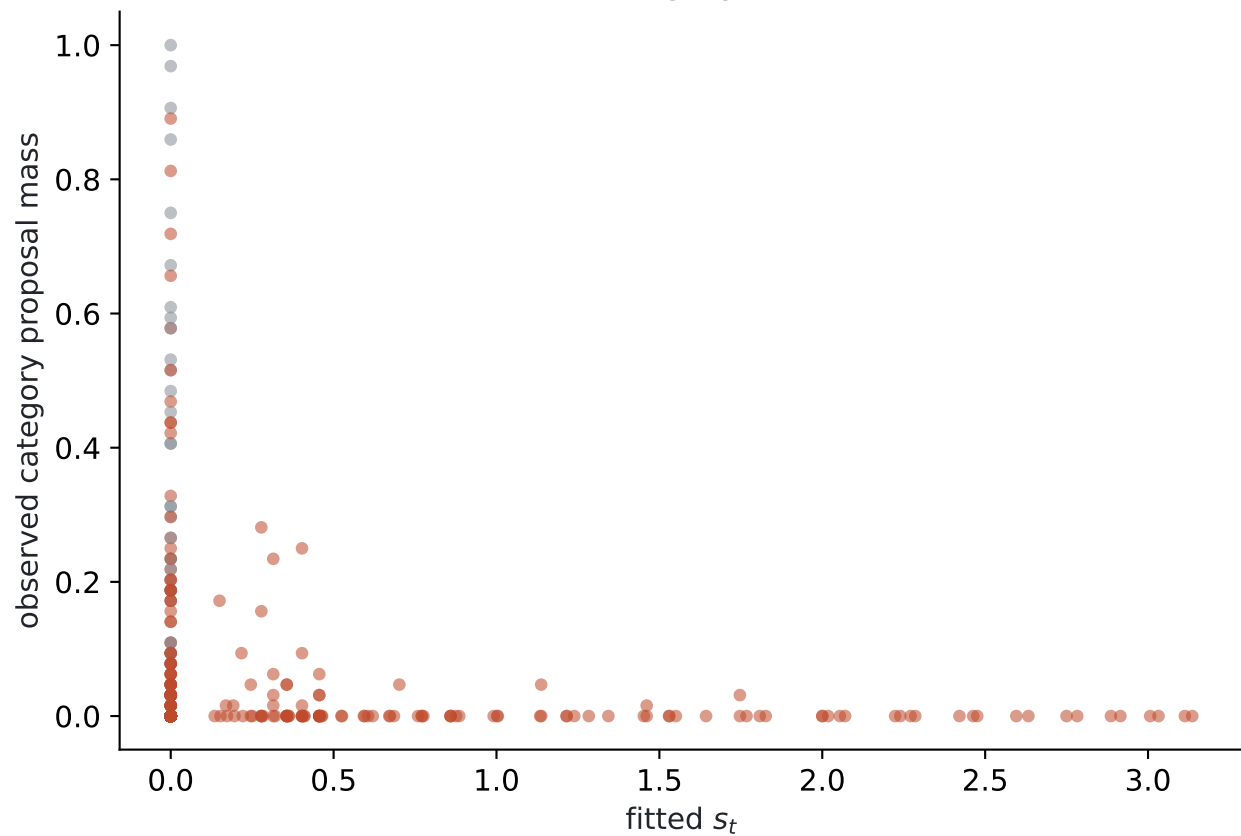


Emission mapping

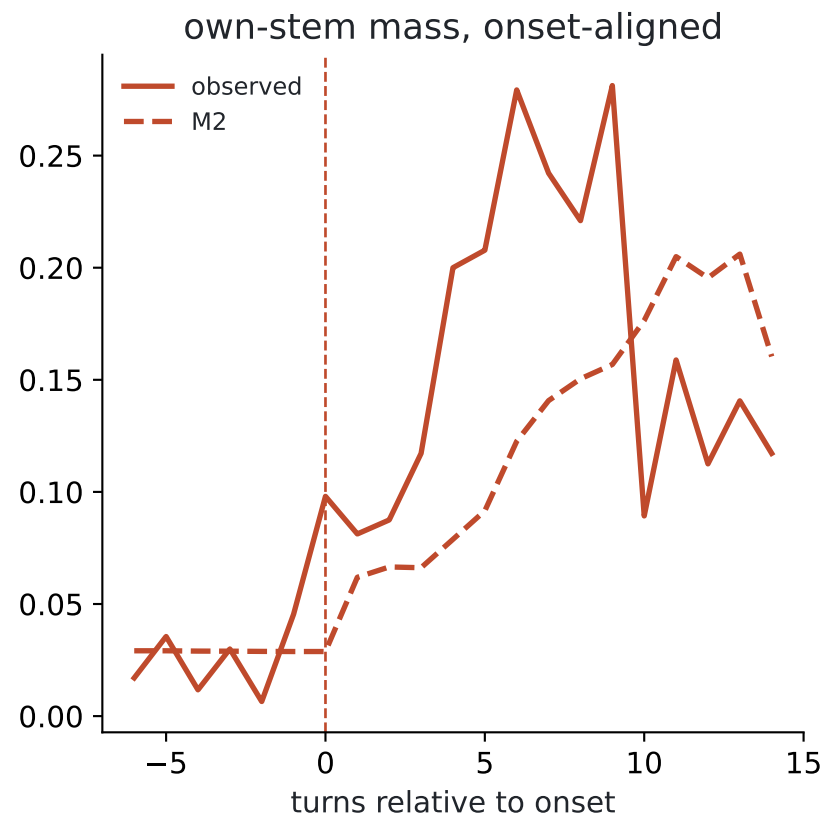
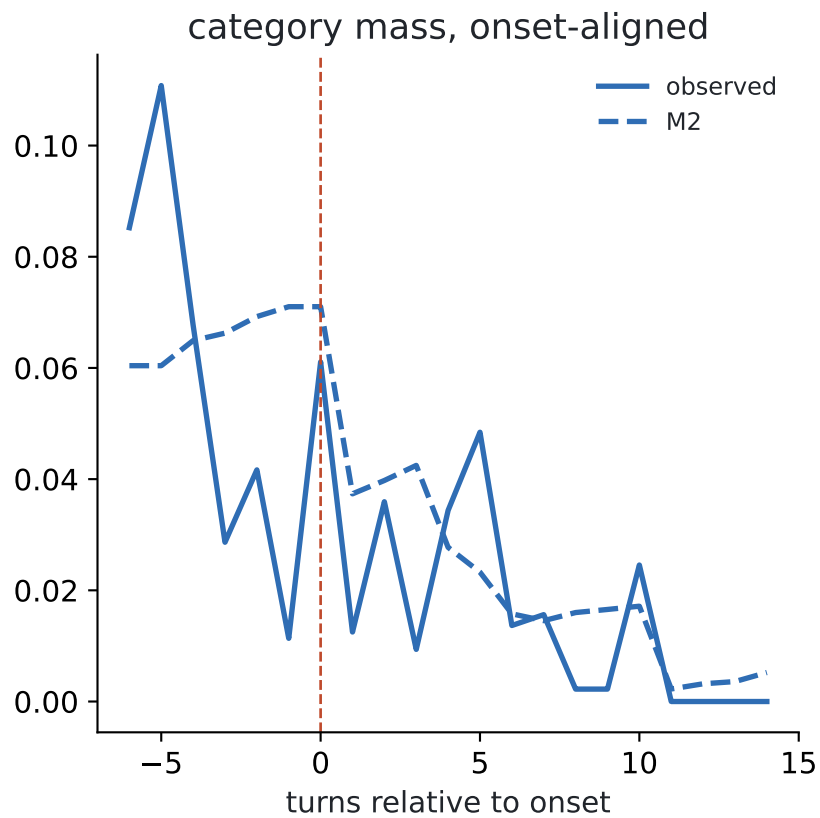
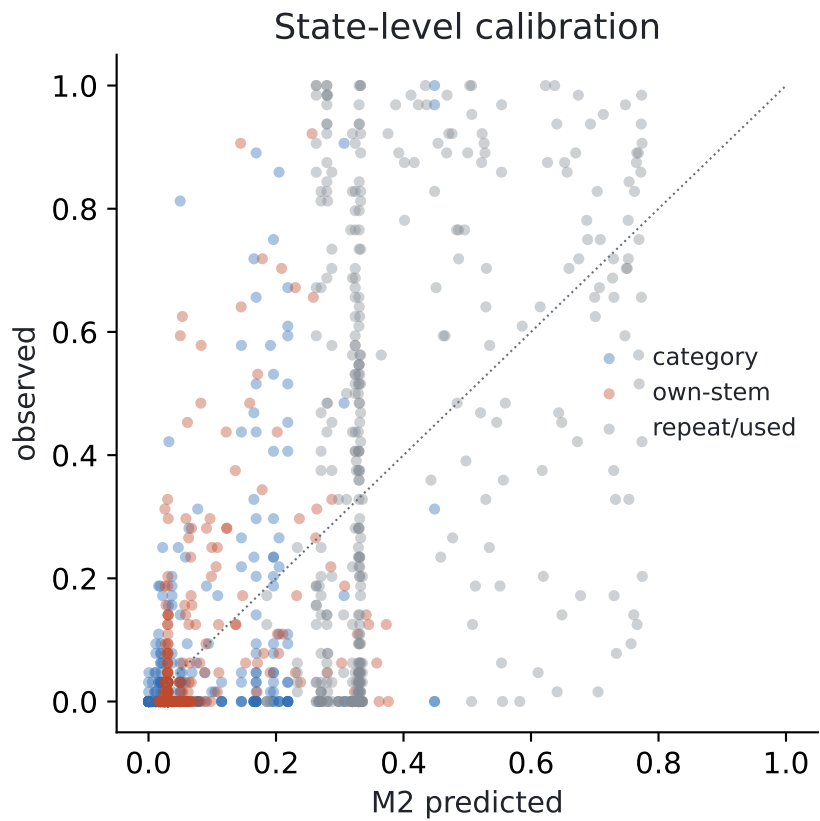
How s reallocates the proposal distribution (at $b = 0, u_g = 0$)



s_t vs observed category mass ($r = -0.16$)

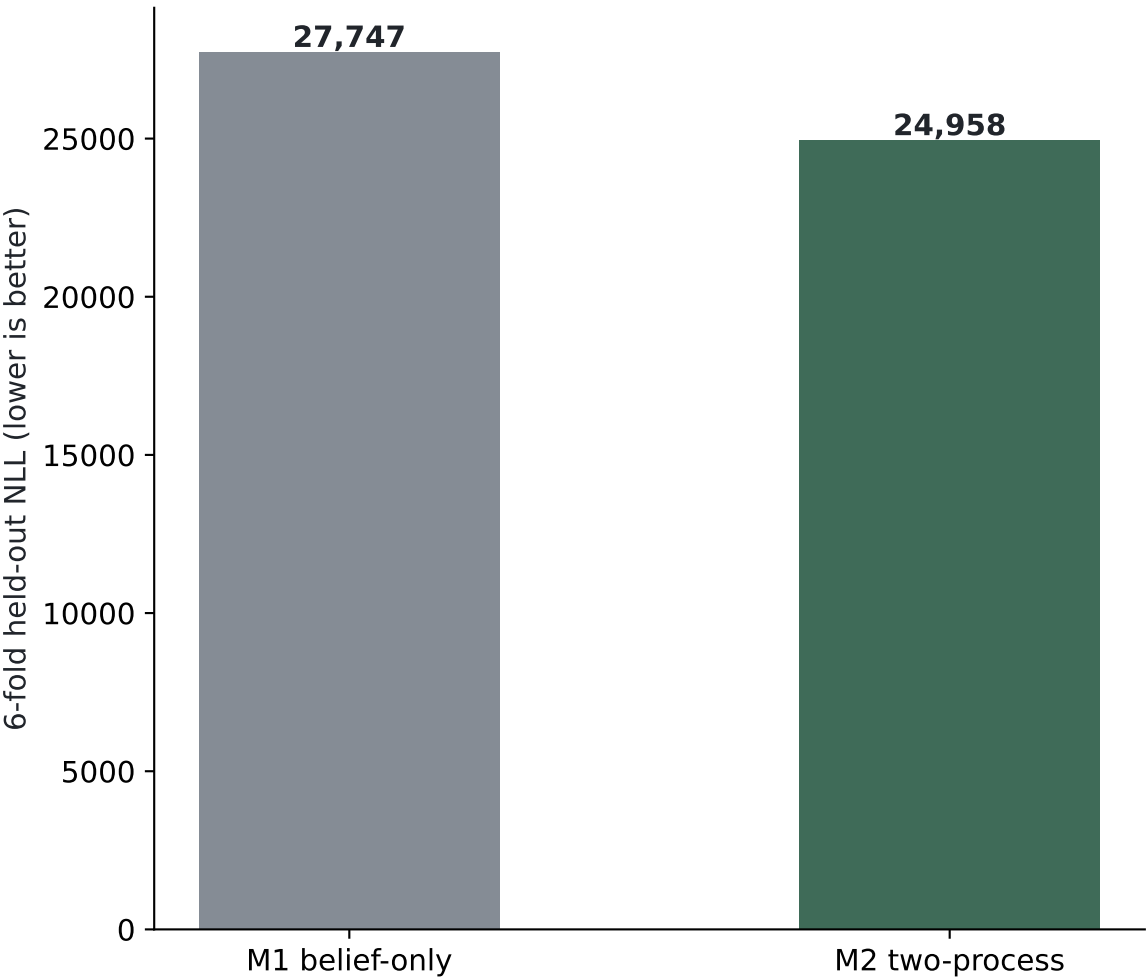


Model fit against data

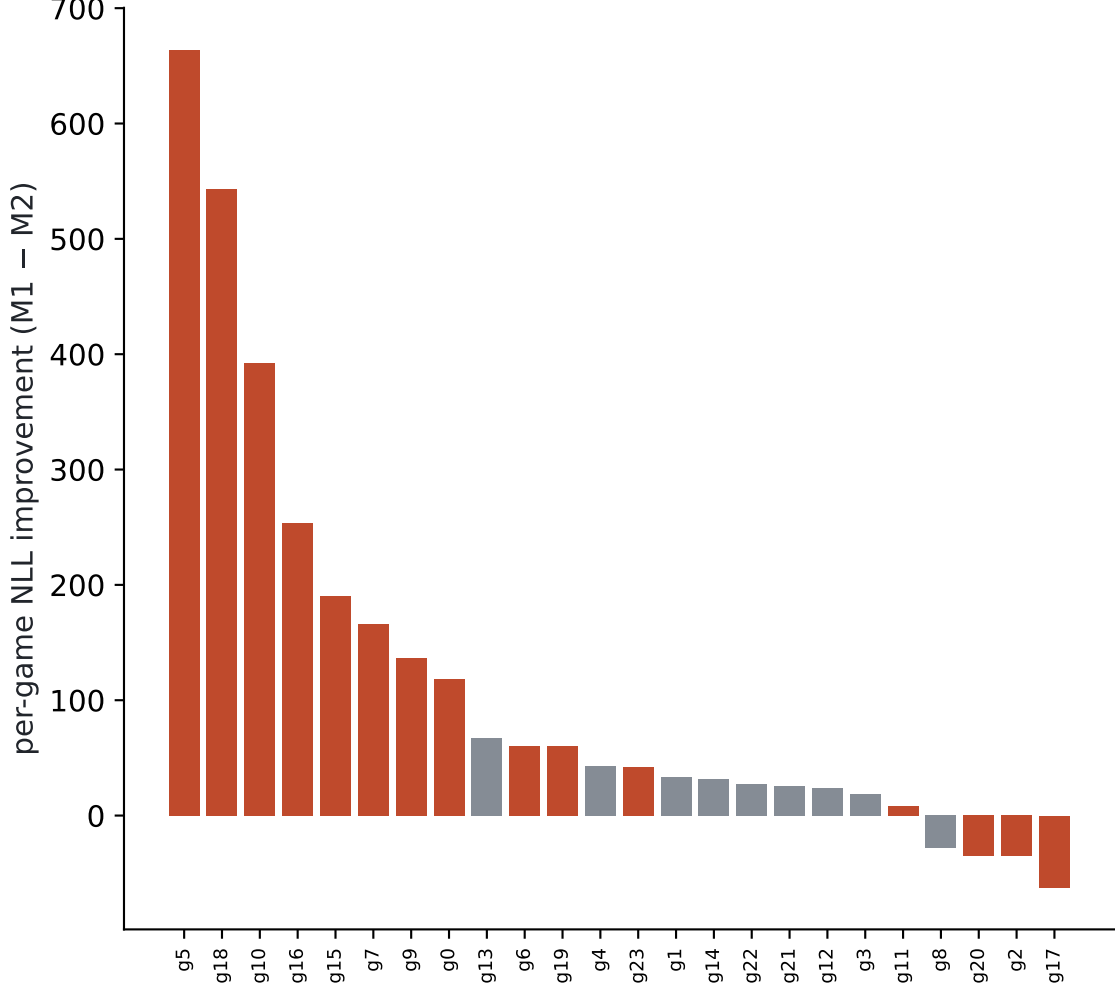


Model comparison

Cross-validated comparison ($\Delta = 2,789$ nats)



Which games need the second process (rust = onset games)



The two-process win concentrates in onset games — the belief-only model has no way to move mass onto the stem/repeat classes when the attractor engages. Mixed-effects AR (companion evidence): $\beta(p_stem)=+3.86$, $\beta(sim)=+1.36$, both $p<1e-10$.